

COURSE TITLE : MEDICAL IMAGING TECHNIQUES
COURSE CODE : 6243
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Radiography and Computed Tomography	19
2	Ultrasound Imaging	18
3	MR Imaging and applications	19
4	Nuclear Imaging and applications	19
TOTAL		75

Course General Outcome :

MODULE	G.O.	ON COMPLETION OF THE STUDY OF THIS MODULE THE STUDENT WILL BE ABLE:
1	1	To Understand the generation , properties of X-ray and x-ray machine
	2	To understand Computed axial tomography and Angiography
2	3	To understand Ultrasound imaging
3	4	To Understand Nuclear Magnetic Resonance Imaging and its applications
4	5	To understand nuclear imaging and its applications
	6	To understand Emission Computed Tomography and PET scanning

G.O.: General Outcome

On completion of the study the student will be able:

MODULE I RADIOGRAPHY AND COMPUTED TOMOGRAPHY

1.1.0 To Understand the generation , properties of X-ray and x-ray machine

- 1.1.1 To define X- Ray
- 1.1.2 To explain the generation of X-ray
- 1.1.3 To list the properties of X- ray
- 1.1.4 To explain the block diagram of X-ray machine

1.2.0 To understand Computed axial tomography and Angiography

- 1.2.1 To explain the principle of computed tomography
- 1.2.2 To explain the system components of CT machine
- 1.2.3 To explain different generations of Computed Tomography
- 1.2.4 To differentiate between CT & X-ray
- 1.2.5 To explain angiography
- 1.2.6 To explain C-ARM and Balloon angioplasty
- 1.2.7 To describe the working of intra aortic balloon pump.

MODULE II ULTRASOUND IMAGING

2.1.0 To Understand Ultrasound imaging

- 2.1.1 To describe ultrasound and its properties
- 2.1.2 To explain the generation of ultrasound
- 2.1.3 To explain different modes of scanning used in ultrasound imaging
- 2.1.4 To explain different types of scanning used in ultrasound imaging
- 2.1.5 To explain Doppler Ultrasound
- 2.1.6 To describe echocardiography
- 2.1.7 To describe echoencephalography
- 2.1.8 To list the applications of ultrasound in medical field

MODULE III MR IMAGING AND APPLICATIONS

3.1.0 To Understand Nuclear Magnetic Resonance Imaging and its applications

- 3.1.1 To explain the principle of NMR imaging
- 3.1.2 To define precession
- 3.1.3 To define Longitudinal and Transverse relaxation time
- 3.1.4 To explain the system components of NMR system
- 3.1.5 To describe important magnets used in MRI
- 3.1.6 To list the applications of MRI

MODULE IV NUCLEAR IMAGING AND APPLICATIONS

4.1.0 To understand nuclear imaging and its applications

- 4.1.1 To explain the concept of nuclear imaging
- 4.1.2 To describe radioactive isotopes used in nuclear imaging
- 4.1.3 To describe radioactive decay
- 4.1.4 To Describe half life period
- 4.1.5 To list the properties of Gamma rays
- 4.1.6 To explain the building components of Gamma camera
- 4.1.7 To list the applications of gamma camera

4.2.0 To understand Emission Computed Tomography PET scanning

- 4.2.1 To explain principle and operation of SPECT scanning
- 4.2.2 To explain principle and operation of PET scanning

CONTENTS

MODULE I GENERATION , PROPERTIES OF X-RAYS AND X-RAY MACHINE

Definition of X- Rays, generation and properties of X-ray- Block diagram of X-ray machine

Computed axial tomography and Angiography

Principle of computed tomography - system components of CT machine- generations of Computed Tomography- Comparison between CT & X-ray - angiography- C-ARM and Balloon angioplasty - intra aortic balloon pump.

MODULE II ULTRASOUND IMAGING

Ultrasound and its properties- Generation of ultrasound - Modes of scanning used in ultrasound imaging- Types of scanning- Doppler Ultrasound- Echocardiography –Echoencephalography-applications of ultrasound in medical field

MODULE III NUCLEAR MAGNETIC RESONANCE IMAGING AND ITS APPLICATIONS

Principle of NMR imaging

Definition of precession, Longitudinal and Transverse relaxation time-System components of NMR system-Magnets used in MRI- Applications of MRI

MODULE IV NUCLEAR IMAGING AND ITS APPLICATIONS

Concept of Nuclear imaging- Radioactive isotopes used in nuclear imaging-Radioactive decay-Half life period-properties of Gamma rays- Building components of Gamma camera- Applications of gamma camera

Emission Computed Tomography PET scanning

Principle and operation of SPECT scanning and PET scanning

TEXT BOOKS

1. Handbook of Biomedical Instrumentation- RS Khandpur- Tata McGraw-Hill Pub., 2006t
2. Physics of Medical Imaging-Save Webb-IOP Publishing Ltd-1988

REFERENCES

1. Fundamental Physics of radiology-Massey & Meredith-Wright, Bristol-1992
2. Principles of computed tomography-A C Kak-IEEE Press New York
3. Physics of diagnostic ultrasound-Peter Fisch-John Wiley & sons-England-1990
4. MRI made easy-H H Schild-Schering A G-2003